

MTH204 (Section A)
End Semester Exam

May 2nd, 2026

2 hours

Total Marks: 80

Name:

Roll No.:

Important Notice: Only a pen and four **A4-sized handwritten cheat sheet** are permitted during the exam. No other materials, electronic devices, or aids are allowed. **No discussion or communication of any kind is permitted. Please return the question paper along with the answer sheet and please do not write anything on the question paper other than your name and the roll number.**

Q1.(12 Marks) For the given Sturm–Liouville problem

$$y'' + 8y' + (\lambda + 16)y = 0,$$

with boundary conditions

$$y(0) = 0, \quad y(\pi) = 0,$$

find the eigenvalues and eigenfunctions. Verify orthogonality with respect to the appropriate weight function.

Q2.(12 Marks) Solve the following system of equations using the method of variation of parameters:

$$\begin{cases} x_1' = x_1 - x_2 + e^{2t}, \\ x_2' = -4x_1 + x_2 + t, \end{cases}$$

with

$$x_1(0) = 0, \quad x_2(0) = 1.$$

Q3.(12 Marks) Consider the PDE

$$U_{xx} + 5U_{xy} + 4U_{yy} = 0.$$

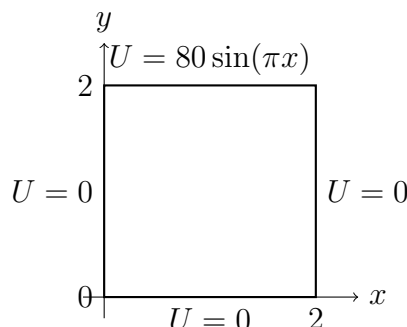
- (a) Determine the type of the above second order PDE.
- (b) Find the characteristic equation and hence obtain a suitable transformation of variables.
- (c) Using the transformation, reduce the given equation to its normal form (show all your work).

(d) Hence obtain the general solution of the equation.

Q4.(12 Marks) A thin square metal plate of side 2 units is shown in the figure below.

The edges of the plate are maintained at the following temperatures:

- The left edge $x = 0$ and the right edge $x = 2$ are kept at 0°C .
- The bottom edge $y = 0$ is kept at 0°C .
- The top edge $y = 2$ is maintained at temperature $80 \sin(\pi x)$.



Assume that the plate has reached a steady-state temperature distribution and that there are no internal heat sources.

Using the method of separation of variables, solve the resulting boundary value problem and obtain the expression for the temperature distribution $U(x, y)$ in the plate.

Q5.(10 Marks) Assuming a solution in the form of trigonometric series, find the solution of

$$y'' + \omega^2 y = r(t),$$

where $|\omega| \neq 0, 1, 2, \dots$ and $r(t) = |t|$ is 2π -periodic for $-\pi \leq t \leq \pi$.

Hint: Use Fourier series.

Q6.(10 Marks) If $P_m(x)$ and $P_n(x)$ are Legendre polynomials of degree m and n respectively, show that

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0 \quad \text{for } m \neq n.$$

Q7.(12 Marks) Let $P(s)$ and $Q(s)$ be two polynomials where the degree of $P(s)$ is less than the degree of $Q(s)$. Assume the zeros of Q are real, distinct, and given by $\alpha_1, \alpha_2, \dots, \alpha_n$. For simplicity, assume that the coefficient of the highest power of s in $Q(s)$ is 1. Show that

$$\mathcal{L}^{-1} \left\{ \frac{P(s)}{Q(s)} \right\} = \sum_{k=1}^n \frac{P(\alpha_k)}{Q'(\alpha_k)} e^{\alpha_k t}.$$

Hint: Consider the partial fraction decomposition of

$$\frac{P(s)}{Q(s)}.$$